MEP/80
MAG

JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 85, NO. B2, PAGES 997–1012, FEBRUARY 10, 1980

Ref 1221

Early Acquisition of Characteristic Magnetization in Red Beds of the Moenkopi Formation (Triassic), Gray Mountain, Arizona

MICHAEL E. PURUCKER,[1] DONALD P. ELSTON, AND EUGENE M. SHOEMAKER

U.S. Geological Survey, Flagstaff, Arizona 86001

Four discrete magnetizations associated with events occurring during deposition, early diagenesis, recent weathering, and cloud to ground lightning have been identified in rocks of the Moenkopi Formation at Gray Mountain, Arizona. A paleomagnetic study of more than 800 samples tied closely to the physical stratigraphy of the Moenkopi indicates that most of the magnetization remaining after partial alternating field and thermal demagnetization was acquired during and shortly after deposition. Directions of this remaining (characteristic) magnetization, in both normally and reversely magnetized rock, are closely similar to directions reported for the Moenkopi elsewhere on the Colorado Plateau. Sampling across the Moenkopi Formation at Gray Mountain has revealed two reversed magnetozones separating three normal magnetozones. Generally sharp boundaries of the magnetozones, their close concordance with physical stratigraphic units, and the rarity of anomalous directions of magnetization within the magnetozones indicate that most of the secondary components of the characteristic magnetization were acquired within 10^4–10^5 years after deposition. A conglomerate test based on siltstone clasts in sandstone shows highly scattered but not completely random directions of magnetization following partial thermal demagnetization. A secondary Triassic normal magnetization appears to be superimposed on some of the clasts, as suggested by the approximate coincidence of the resultant magnetic vector from the clasts with a Triassic normal direction. A fold test based on samples from highly deformed silty clay stone and sandstone in an intraformational landslide shows that these rocks acquired most of their stable magnetization before folding. A component of Triassic normal magnetization acquired after deformation also occurs in some of these rocks.

INTRODUCTION

Almost from the start of modern paleomagnetic research that flourished after the development by Blackett of a sensitive astatic magnetometer, paleomagnetists have devoted considerable attention to the paleomagnetism of hematite-bearing red beds. This interest in red beds stemmed, in part, from the fact that their magnetization could be easily measured and, in part, from the stability of this magnetization. Only recently, however, with the invention of techniques of chemical demagnetization [*Collinson*, 1967; *Roy and Park*, 1972] and a landmark study by *Collinson* [1974], has a corresponding effort gone into understanding the magnetic properties of hematite-bearing sedimentary rocks and the time of acquisition of magnetization in these rocks.

In this paper we report several lines of evidence which bear on the time of acquisition of magnetization in red beds of the Moenkopi Formation near Gray Mountain, Arizona. More than one generation of hematite has been recognized in these strata [*Walker and Larson*, 1976], and the rocks have been described as magnetically complex and unsuitable for investigations of magnetostratigraphy [*Larson et al.*, 1976]. Our results indicate that, providing care is taken to sample the freshest outcrops and the effects of lightning are recognized, the Moenkopi Formation is well suited for magnetostratigraphic studies. In fact, four well-defined magnetozones found at Gray Mountain have been traced by us more than 350 km along the Moenkopi outcrop belt in Arizona and southwestern Utah. A companion paper [*Elston and Purucker*, 1979] summarizes evidence for the detrital origin of a major part of the stable magnetization in beds of the Moenkopi Formation at Gray Mountain.

[1] Now at Department of Geological and Geophysical Sciences, Princeton University, Princeton, New Jersey 08540.

This paper is not subject to U.S. copyright. Published in 1980 by the American Geophysical Union.

Paper number 9B1423.

PHYSICAL STRATIGRAPHY

The Moenkopi Formation of Triassic age [*Ward*, 1901; *McKee*, 1954; *Stewart et al.*, 1972] is exposed in a broad outcrop belt across the southern Colorado Plateau (Figure 1). *McKee* [1954, p. 19] recognized three members of the Moenkopi in the Little Colorado River Valley. In ascending stratigraphic order they are the Lower Triassic Wupatki Member and the Lower and Middle(?) Triassic Moqui and Holbrook Members (Figure 2).

The Wupatki Member rests unconformably on the Kaibab Limestone of Permian age and is composed dominantly of pale reddish-brown, horizontally laminated and ripple-laminated siltstone that weathers to slopes. A widely distributed, ledge-forming sandstone occurs at the top of the Wupatki Member and has been called the lower massive sandstone by *McKee* [1954]. The lower massive sandstone is identical with the 'Wupatki Sandstone' of *Larson et al.* [1976]. We use here McKee's nomenclature. A special investigation of the lower massive sandstone has been conducted by *Baldwin* [1973], who distinguished four units within the lower massive sandstone on the basis of bedding characteristics. Three informal units of the lower massive sandstone have been recognized in this investigation at Gray Mountain, Arizona, which correspond to Baldwin's units 2, 3, and 4. The lower and upper units are cliff-forming, very fine grained, parallel- and cross-laminated, very pale reddish-brown sandstone. The middle unit, corresponding to Baldwin's unit 3, is discontinuous and is recognized locally over a distance of 10 km or more north of Gray Mountain. Where sampled by us it consists of very fine grained sandstone, silty sandstone, and silty clay stone. At most places the middle unit is absent, and the lower and upper units are in contact along a laterally persistent diastem. In the area studied, erosion of the lower unit locally produced a swale in which beds of the middle unit were deposited. Deposition of the middle unit was followed, in turn, by erosion and by deposition of the upper unit. In contrast to McKee we place the top

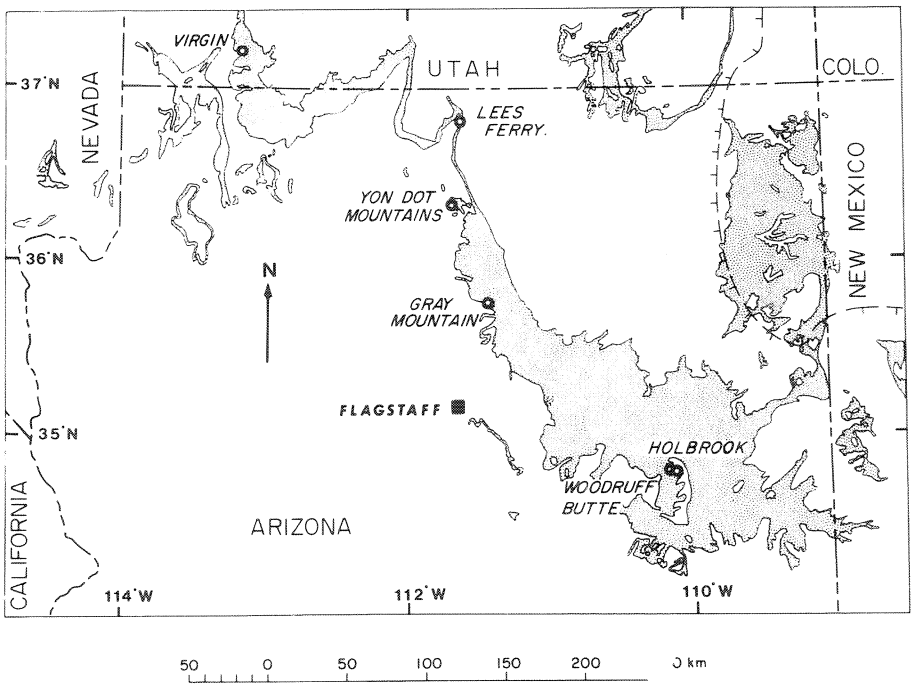


Fig. 1. Map showing locations of stratigraphic sections of the Moenkopi Formation in Arizona and southwestern Utah that have been sampled by the authors for paleomagnetic investigations. Stippled areas, modified from *Stewart et al.* [1972], are outcrops of Triassic strata. The Moenkopi Formation is present west and south of the dashed, hachured line. Open circles are locations where the entire Moenkopi Formation has been sampled for magnetic studies.

of the Wupatki Member at the top of the lower massive sandstone rather than at the base of the lowermost gypsiferous strata in the overlying Moqui Member. This permits a less equivocal tracing of the Wupatki-Moqui boundary.

The Moqui Member is composed dominantly of slope-forming, pale reddish-brown, horizontally laminated siltstone with minor interstratified white gypsum. The gypsum occurs as nodules, as thin horizontal beds, and as thin seams cross-cutting the stratification. In contrast to McKee we place the top of the Moqui Member at the base of the lowest persistent ledge-forming sandstone of the overlying Holbrook Member and not at the top of the highest interstratified gypsum.

The Holbrook Member consists dominantly of interstratified and interfingering layers of pale reddish-brown sandstone and siltstone. At the Gray Mountain section the contact of the Holbrook with the Moqui Member is a local unconformity. Elsewhere the Holbrook Member interfingers with the Moqui Member. The Holbrook is overlain unconformably by the Chinle Formation.

Most of the investigation reported here was carried out at an exceptionally well exposed stratigraphic section 75 km north of Flagstaff, Arizona, and 7–9 km northwest of the trading post at Gray Mountain. The Moenkopi at this section is about 120 m thick. The exposures occur south and east of the intersection of Tappan and Cedar washes in Sec.14,T.28N.,R8E., Coconino Point SE 7½′ quadrangle and in Sec.24,T.28N.,R8E., Cameron SW 7½′ quadrangle (latitude 35°48′N, longitude 111°30′W).

Magnetic Studies

Field and Laboratory Procedures

More than 800 cores, 2.5 cm in diameter and 5–15 cm long, were collected with a portable gasoline-powered core drill and with orientation equipment similar to that described by *Doell and Cox* [1965]. Stratigraphic separations between cores taken through the entire formation ranged from 3 to 76 cm. The azimuth and plunge of the individual cores, each of which is classed as a sample if it occupies a discrete statigraphic horizon, were measured to the nearest ½° with an estimated accuracy of 1°–2°. One to four specimens were cut from each core. Remanent magnetization was measured with a superconducting rock magnetometer that has a background noise of $\sim 1 \times 10^{-7}$ emu or less and also with an air-driven spinner magnetometer that has a noise level of $\sim 3 \times 10^{-7}$ emu.

All specimens were subjected to thermal demagnetization in air in one of two ovens. Oven I uses Helmholtz coils to control the magnetic field in the vicinity of the specimens and heats and cools the specimens in a magnetic field that is generally less than 50 γ. Oven II uses both Helmholtz coils and multiple mu-metal shields to control the magnetic field in the vicinity of the specimens, and the magnetic field in their vicinity is almost always less than 15 γ. The only difference noted in the results obtained from the two ovens was a tendency for oven I to indicate temperatures slightly higher than the true temperature by about 10°–20°C in the 650°–670°C range.

A variety of thermal demagnetization procedures were used for different sets of specimens. Almost all specimens were subjected to thermal demagnetization at 550°C. Approximately one third of the specimens were subjected to progressive thermal demagnetization in steps ranging from 200° to 670°C. Groups of 10–40 specimens were demagnetized together during the individual heating steps. In detailed magnetic studies for conglomerate and fold tests, representatives of the different rock types were included in each group; the specimens were loaded in nonsystematic positions and orientations, and the positions and orientations were recorded for each heating step. The mix of different rock types and orientations allowed components of magnetization that might accidentally be superimposed on a specimen or group of specimens during thermal demagnetization to be recognized. Two specimens acquired a spurious magnetization during cooling from 500°C in oven I. These two specimens responded to further thermal

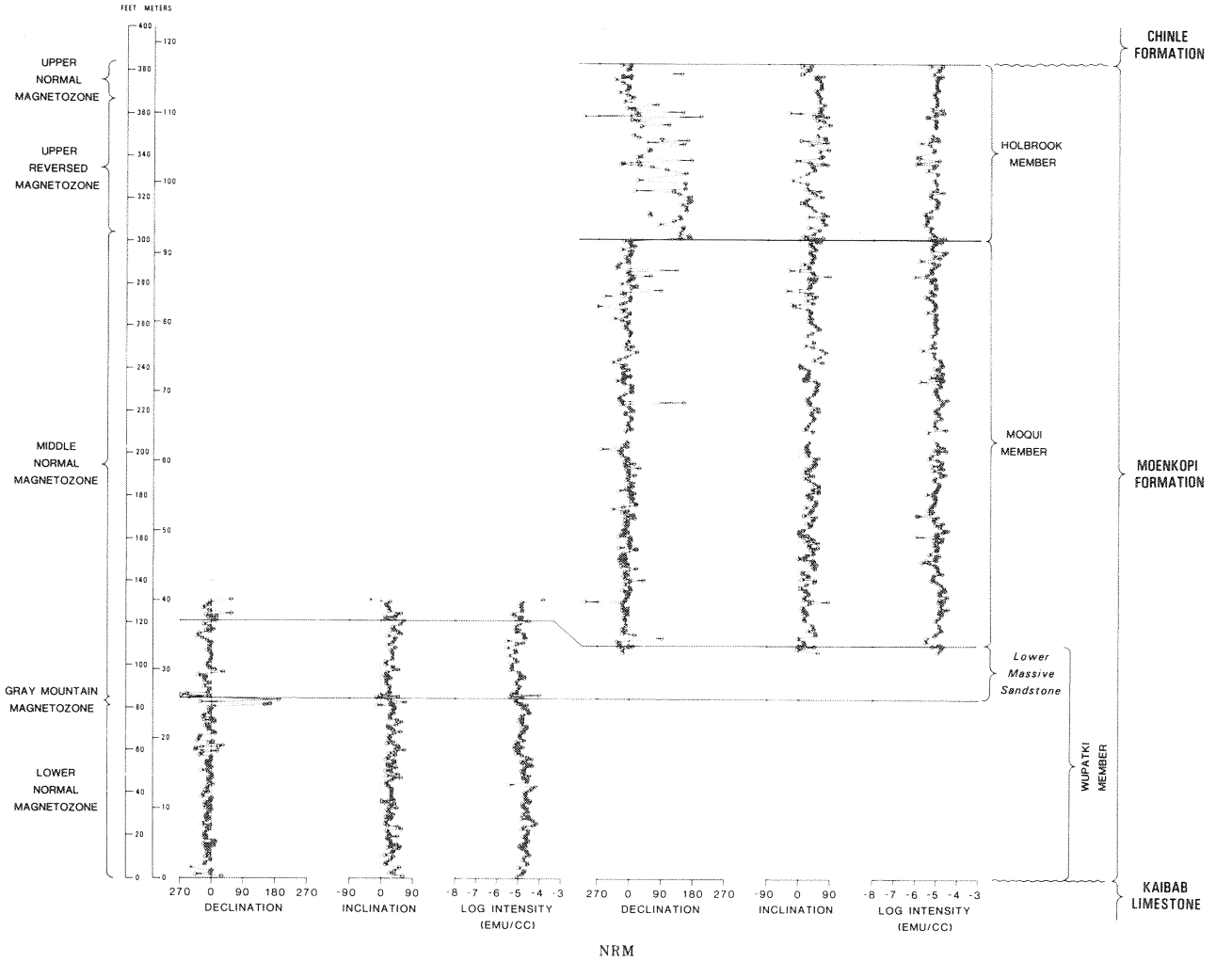


Fig. 2*a*. Paleomagnetic directions from the Moenkopi Formation at the Gray Mountain, Arizona, section for natural remanent magnetization (NRM). The section was sampled at two locations approximately 1 km apart; correlation between these two locations was made by using the base of the lower massive sandstone as a datum. Names of physical stratigraphic units are to the right; informal magnetostratigraphic names are to the left. Log intensity is given as electromagnetic units per cubic centimeter.

demagnetization and ultimately displayed directions typical of the Moenkopi Formation after demagnetization at 650°C.

Twenty-five samples reported here displayed anomalously strong remanent intensities, a characteristic indicative of lightning [*Purucker*, 1974]. These samples were subjected to alternating field (af) demagnetization with an af demagnetizer of commercial design to remove the isothermal remanent magnetization (IRM) component [*Cox*, 1961; *Graham*, 1961], a procedure that had been carried out in earlier investigations on the effects of lightning-produced magnetization in the lower massive sandstone at Gray Mountain [*Purucker*, 1974]. It is important that the effects of lighting be recognized before thermal demagnetization is attempted. If the IRM is not removed before thermal demagnetization, heating seems to 'set' the anomalous component, and it becomes impossible to remove, even if af demagnetization is subsequently performed.

Magnetic Zonation of the Moenkopi Formation at Gray Mountain

The entire Moenkopi Formation at Gray Mountain was sampled at a relatively uniform stratigraphic interval averaging ~0.3 m (Figure 2). The natural remanent magnetization (NRM) was measured for 437 samples from this section; 425 samples survived partial thermal demagnetization to 550°C. Directions of NRM from samples collected at the outcrop commonly show the effects of weathering, which is revealed by secondary components of magnetization that approximately parallel the present-day axial dipole field direction. In a given magnetozone the NRM directions of the samples generally are distributed in a broad swath or 'streak' (Figure 3) that extends from the vicinity of the present-day axial dipole field direction to the vicinity of a characteristic direction of magnetization, defined as a stable direction of magnetization found after partial demagnetization of the specimens to temperatures of 500°C and higher. In most specimens the present field component of magnetization is largely removed by demagnetization to 400°C.

Two directions of characteristic magnetization of nearly opposite polarity were found after partial thermal demagnetization of rocks from the stratigraphic section of the Moenkopi at Gray Mountain and from detailed sampling of one reversed polarity magnetozone (Table 1). The means of these two directions are 171° apart. The shortest great circle arc between them passes near the present-day axial dipole field direction. This indicates that thermal demagnetization did not com-

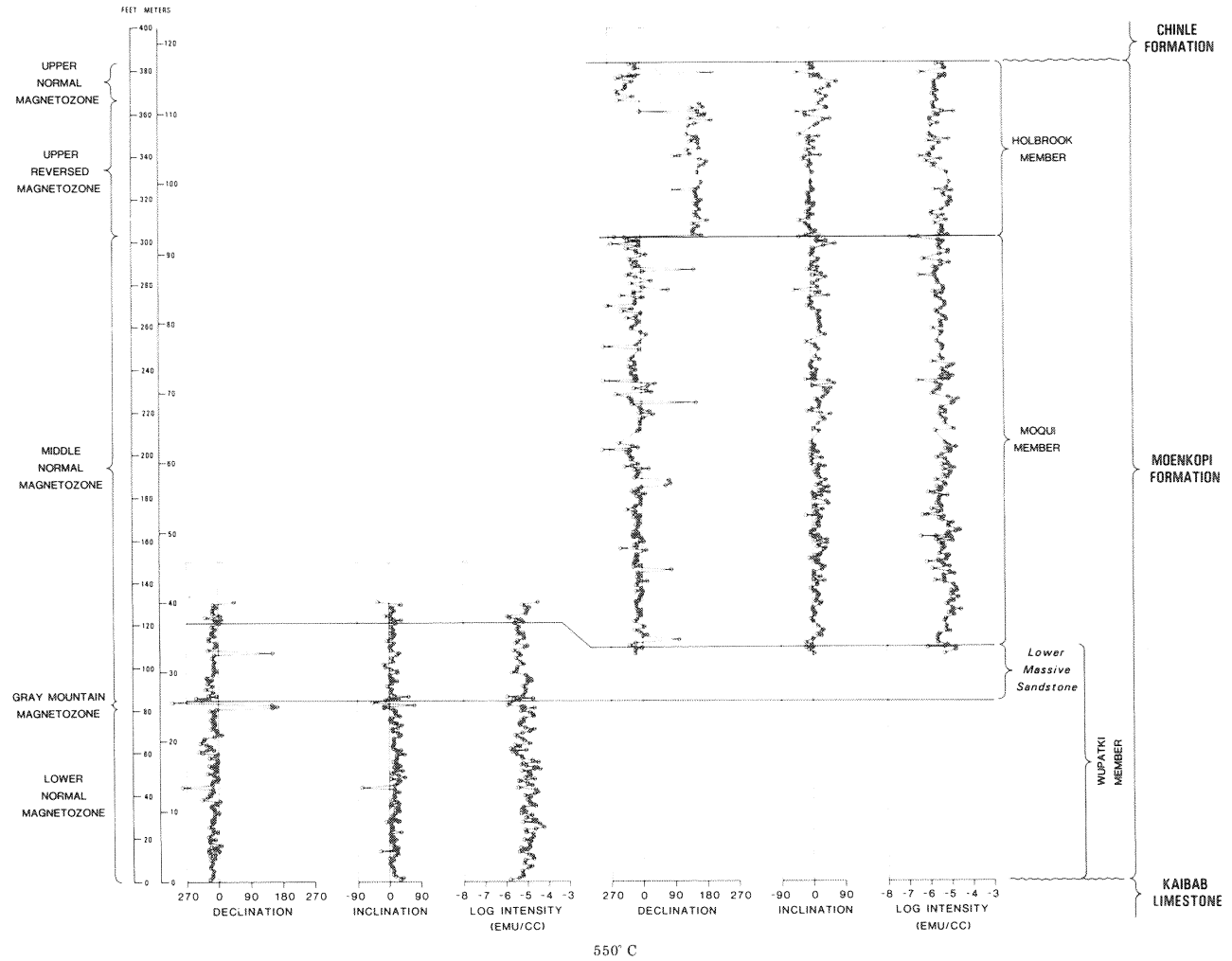


Fig. 2*b*. Same as Figure 2*a* but for remanent magnetization after thermal demagnetization to 550°C.

pletely remove in all specimens secondary components related to weathering in the present field. The largest effect is found in reversely magnetized rocks from near the top of the Moenkopi, rocks that appear to have been exposed longest to weathering. If only the directions from a reversed magnetozone low in the Moenkopi are compared with the normally magnetized directions, the two mean characteristic directions are 173° apart. We interpret that the directions of characteristic magnetization of the Moenkopi reflect directions of the magnetic field during Moenkopi deposition but are slightly biased by residual components of late secondary magnetization.

The directions found at Gray Mountain are closely similar to directions of characteristic magnetization reported from the Moenkopi and equivalent rocks elsewhere on the Colorado Plateau and adjacent central Rocky Mountains [*Helsley*, 1969; *Helsley and Steiner*, 1974; *Baag and Helsley*, 1974; *Christensen and Helsley*, 1974]. For example, *Helsley and Steiner* [1974] report a pole at 55°N, 103°E (dp 1.6°, dm 3.2°) for the Moenkopi Formation of western Colorado. By using the mean direction of magnetization obtained from all normally polarized rocks (demagnetized to 550°C) of the Moenkopi at Gray Mountain, the mean magnetic pole lies at 58°N, 101°E (dp 1.3°, dm 2.5°), within 3° of the pole calculated by Helsley and Steiner. This close correspondence of magnetic directions in different parts of the same formation separated by more than 300 km suggests an early acquisition of the characteristic magnetization. Paleomagnetic poles for the Moenkopi calculated from magnetic directions following demagnetization to 550°C are also similar to other Triassic and Late Triassic to Early Jurassic paleomagnetic poles reported from North America [*Irving and Banks*, 1961; *Opdyke*, 1961; *Beck*, 1965; *Larochelle and Currie*, 1967; *Robertson*, 1967]. These poles represent directions obtained from hematite-bearing sedimentary rocks, from basaltic flows and related intrusives, and from rocks crystallized from an impact melt. Directions of characteristic magnetization close to an average Triassic normal polarity direction are here referred to as normal, and those close to an average Triassic reverse polarity direction are here referred to as reversed.

Rocks with normal characteristic magnetization and those with reversed characteristic magnetization are rather cleanly segregated in discrete stratigraphic intervals at the Gray Mountain Moenkopi section (Figure 2*b*). At the base of the section is a 25-m-thick sequence of beds that is normally magnetized. Only one sample of 104 collected in this sequence displayed a clearly anomalous direction of magnetization. This sequence is here referred to as the lower normal magnetozone. Above the lower normal magnetozone is a sequence of beds, ranging from 1 to 3 m in thickness, that is reversely magnetized; this sequence is called the Gray Mountain magnetozone, discovered from five samples whose directions of mag-

netization are shown in Figure 2. The magnetic characteristics of this zone at Gray Mountain were subsequently studied from about 150 additional samples. The Gray Mountain magnetozone is succeeded by about 70 m of normally magnetized beds, here called the middle normal magnetozone. Thirteen of 245 samples from this zone display anomalous directions. Above the middle normal magnetozone is a 19-m-thick sequence of reversely magnetized beds, the upper reversed magnetozone. Two of 53 samples display anomalous directions. At the top of the Moenkopi section is the upper normal magnetozone, a 6-m-thick sequence of normally magnetized beds from which one sample of 18 displays an anomalous direction. Study of the sites from which the anomalous samples came has shown that many anomalous samples were collected from IRM anomalies produced by lightning. One anomalous sample came from a zone of weathering at the top of the Moenkopi that had been developed before deposition of the Chinle Formation. Other anomalously magnetized samples were found to be magnetically unstable after thermal demagnetization.

By means of five sections of the Moenkopi Formation (Figure 1) sampled in the same way as the section at Gray Mountain, four of the magnetozones at Gray Mountain have been found to be of regional extent. They have been traced eastward to the vicinity of Holbrook, Arizona, northward to Lees Ferry, Arizona, and northwestward to Virgin, Utah. A detailed study was made of the Gray Mountain magnetozone by means of 13 sections across the region, all of which were sampled on close stratigraphic spacing. The top of the Gray Mountain magnetozone was found to lie within 2 m of the base of the lower massive sandstone of McKee over a distance of 400 km [*Shoemaker and Purucker,* 1974]. At Gray Mountain the base of the upper reversed magnetozone is, at one place, about 0.3 m above the locally unconformable base of the Holbrook Member and, at another place, slightly below the base. At other sections the base of the upper reversed mag-

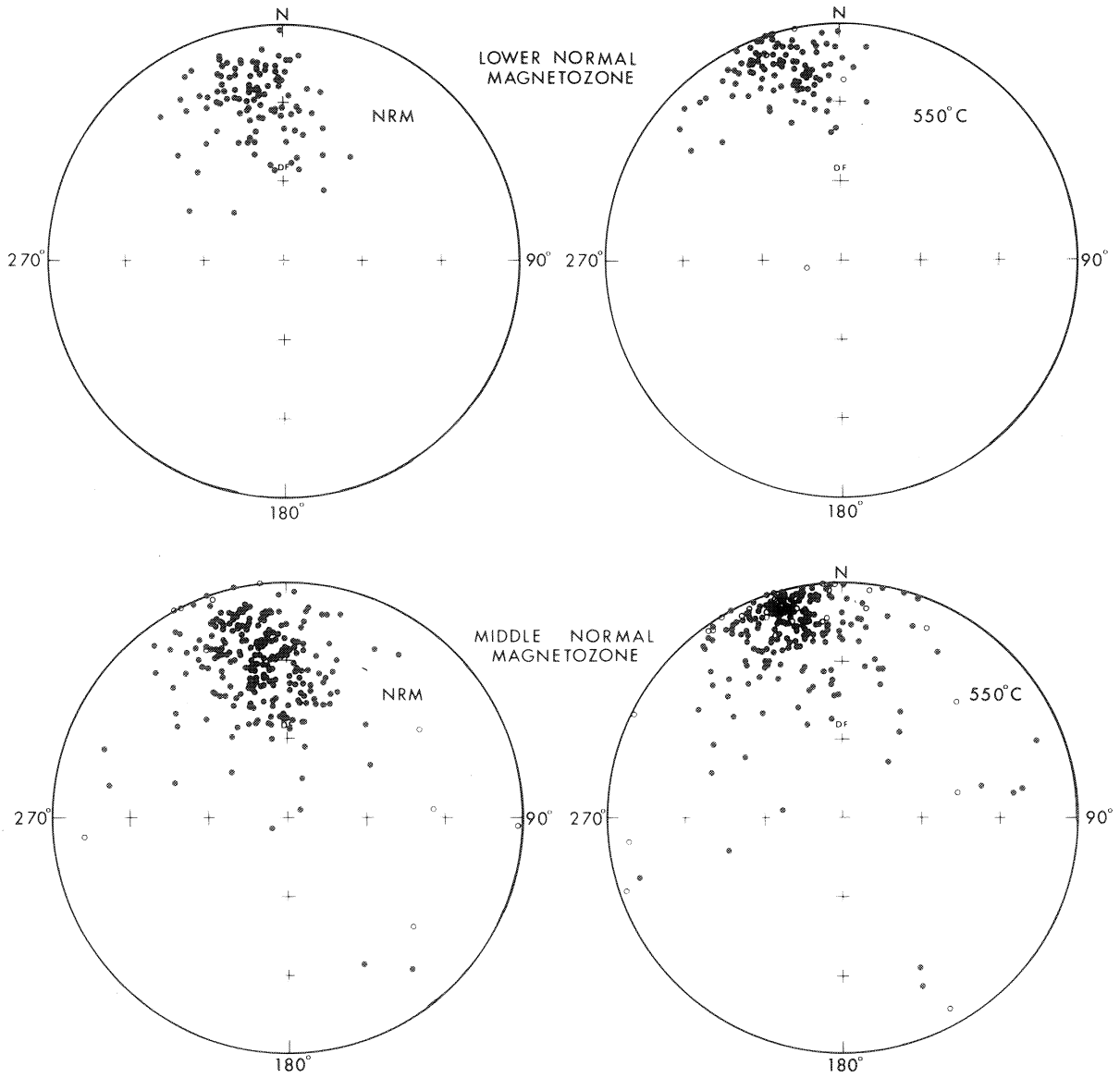


Fig. 3*a*. Equidistant azimuthal projections of directions of magnetization for the Moenkopi Formation at Gray Mountain, Arizona, for NRM and after partial thermal demagnetization for lower and middle normal magnetozones. Solid symbols are on the lower hemisphere (positive inclination). Open symbols are on the upper hemisphere (negative inclination). DF denotes the direction corresponding to a present axial dipole field.

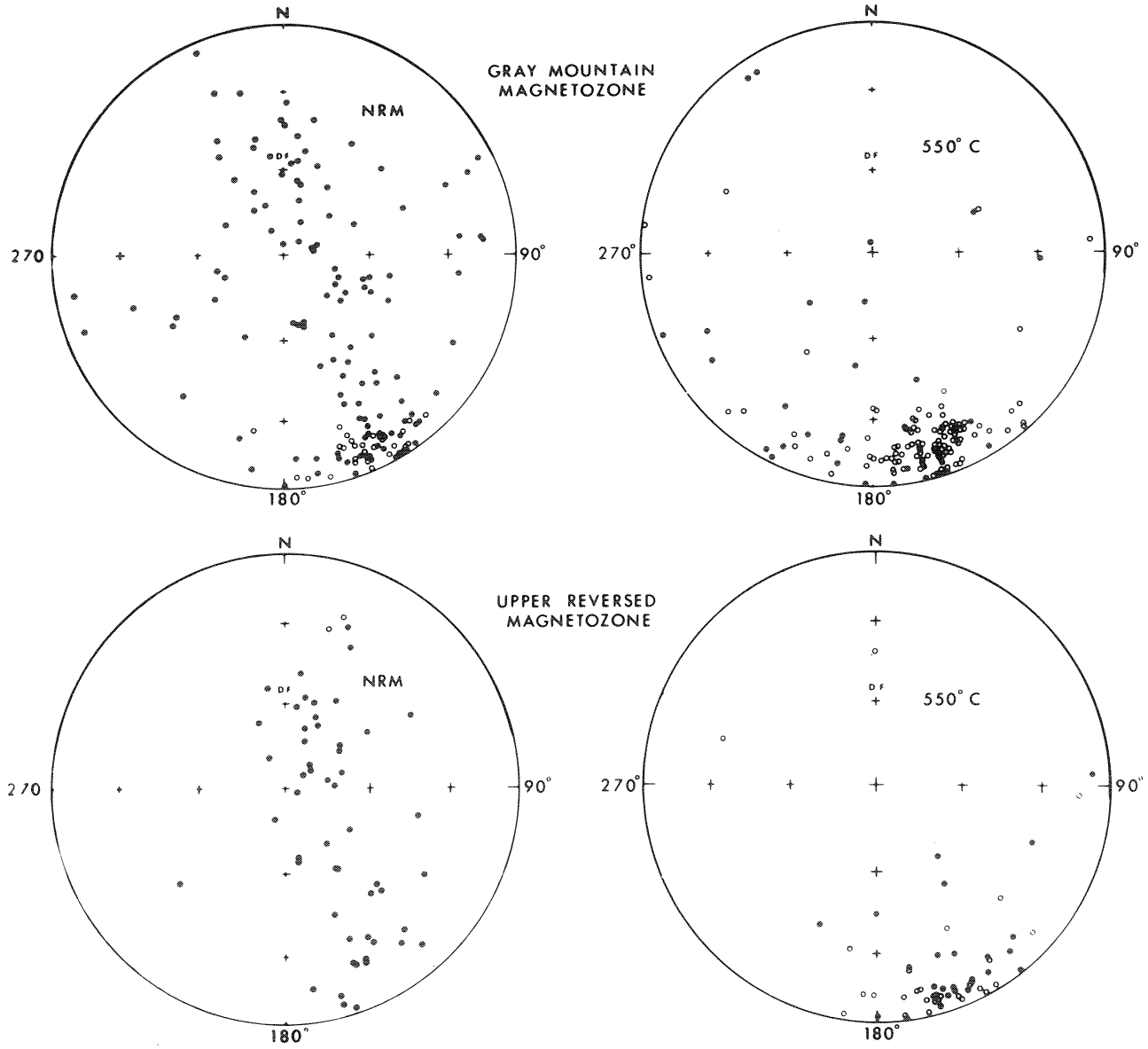


Fig. 3*b*. Same as Figure 3*a* but for Gray Mountain reversed magnetozone and upper reversed magnetozone.

netozone is found either in the Moqui Member a short distance below the contact with the Holbrook or within the Holbrook a short distance above the contact. The upper normal magnetozone has not been identified with certainty at any of the other sections; at most places it appears to have been lost by pre-Chinle erosion. Detailed discussions of the regional relationships and of the paleomagnetic data for the individual sections are to be the subjects of other reports.

The existence of several discrete sequences of beds in the Moenkopi section at Gray Mountain, each with a consistent

TABLE 1. Mean Directions of Magnetization From Paleomagnetic Samples Shown in Figures 2 and 4 Before and After Partial Thermal Demagnetization to 550°C

	$\overline{D}$	$\overline{I}$	k	R	α_{95}	N
		Normal Polarity				
NRM	350.4	32.8	12	376.0	2.1°	411
550°C	343.3	15.4	10	358.6	2.4°	401
		Reversed Polarity				
NRM (all)	144.1	49.1	2.6	126.8	7.8°	206
550°C (all)	166.2	−6.7	5.5	161.9	4.7°	198
550°C (Gray Mountain magnetozone)	168.1	−9.5	6.4	115.6	5.2°	137

NRM is natural remanent magnetization; $\overline{D}$ is mean declination; $\overline{I}$ is mean inclination; k is *Fisher*'s [1953] precision parameter $(N-1)/(N-R)$, where R is the resultant of N directions of magnetization, expressing the scatter of the data points on a spherical surface; α_{95} is the radius of the cone around the mean direction drawn at the 95% confidence level; and N is the number of samples.

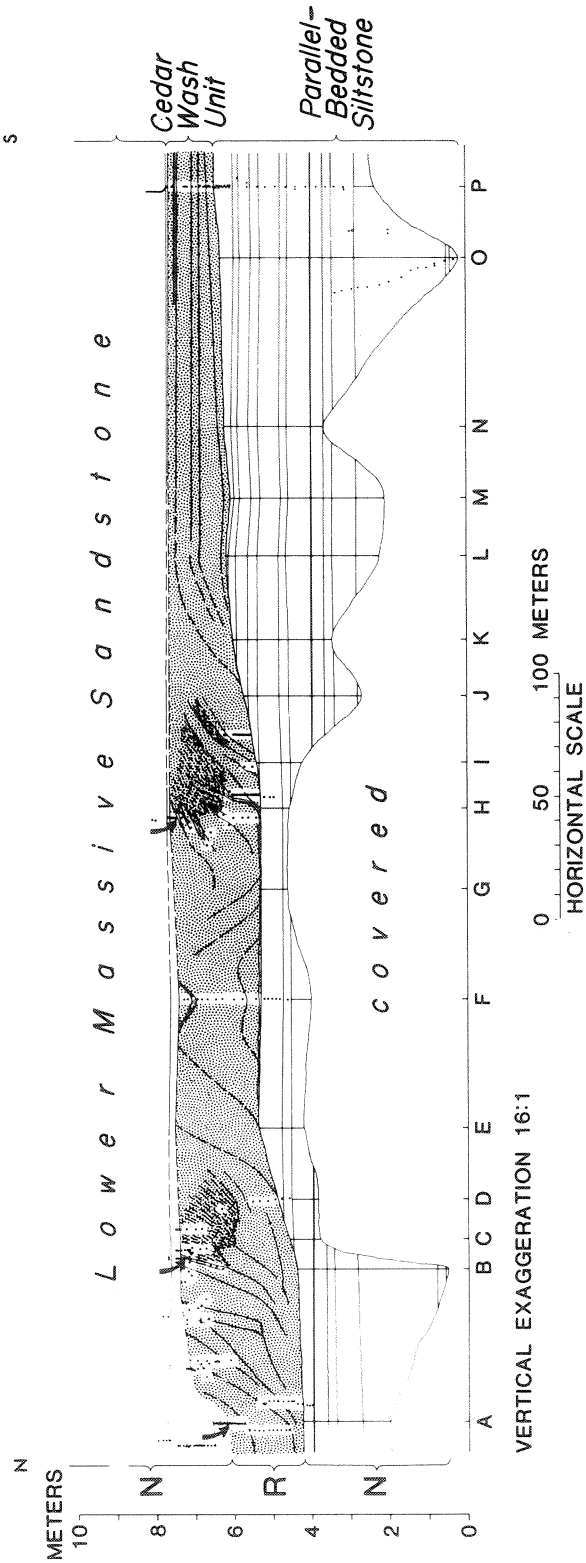


Fig. 4. Correlation of rocks in the cliff face on the east side of north-trending Cedar Wash near Gray Mountain, Arizona, showing the relationship of Gray Mountain magnetozone (stippled) to details of the physical stratigraphy. Informal stratigraphic names are to the right, and the polarity zonation is to the left. The vertical lines above letters A–P show the locations of measured stratigraphic sections. Subhorizontal and inclined lines are prominent bedding planes; these lines are solid where rocks are well exposed and dashed where rocks are incompletely exposed. Closely spaced dashed lines in the Cedar Wash unit indicate shaley beds. Small dots on and near the measured sections represent positions of individual paleomagnetic cores. The arrows near the top of the stippled zone indicate sites of anomalous characteristic magnetization.

polarity of magnetization (with ~4% anomalous directions), the lateral continuity of these sequences or magnetozones, and the general concordance of the magnetozones with the physical stratigraphy on a regional scale together indicate that the characteristic magnetization of the Moenkopi at Gray Mountain was mostly acquired contemporaneously or nearly contemporaneously with deposition. The boundaries between the magnetozones are sharp (i.e., less than the mean sample stratigraphic separation of ~0.3 m) at most places that we have sampled them at Gray Mountain. One of these boundaries occurs at a diastem (local unconformity), and two occur within apparently conformable sequences of beds. The relatively sharp boundaries of the magnetozones observed at Gray Mountain are interpreted by us as corresponding to reversals in polarity of the ancient geomagnetic field during the course of deposition of the Moenkopi.

In western Colorado, 300 m of Moenkopi beds deposited over an interval of time that probably corresponds to the Early Triassic [*Shoemaker and Newman,* 1959], an interval that, very roughly, may have been about 10 million years in duration. The mean rate of deposition in the Moenkopi in western Colorado thus may have been of the order of 3×10^{-5} m yr^{-1}. If the Moenkopi at Gray Mountain, Arizona, was deposited at a similar mean rate, the average sample interval corresponds to a mean time interval of the order of 10^4 years. This is a time long enough to span a polarity reversal of the geomagnetic field [*Fuller et al.,* 1979]. The precision with which a given magnetozone boundary records the time of reversal of the geomagnetic field is, of course, limited by the completeness of the stratigraphic record. Where there is a hiatus, as at the base of the Gray Mountain magnetozone, several tens or even hundreds of thousands of years of earth history may not be represented. Where magnetozone boundaries occur at diastems or local unconformities, they have invariably proven to be as sharp as can be resolved with our standard method of sampling.

Gray Mountain Magnetozone

A special study of the Gray Mountain magnetozone was carried out at Gray Mountain, Arizona, to examine the local relationships of the magnetozone boundaries to the physical stratigraphy. Superb cliff exposures of these boundaries and the entire magnetozone along the wall of a small canyon cut by Cedar Wash make it possible to sample the section of interest in almost any desired detail.

The Gray Mountain magnetozone (Figure 4) occurs within the Wupatki Member of the Moenkopi at Gray Mountain, a short distance below the lower massive sandstone. At this section the lowermost 25 m of the Wupatki Member consists of a sequence of parallel-bedded siltstone with rare thin interbedded fine to very fine grained sandstone. Bedding in the siltstone is extremely uniform. Individual depositional units at the top of the sequence, ranging in thickness from 3 to 60 cm, have been traced for more than 1 km; only minor changes in thickness and lithology were found in that distance. Salt casts and abundant desiccation cracks found in this sequence, together with the uniformity of bedding, suggest that it is a supratidal deposit. The parallel-bedded siltstone is overlain by a unit of fine to very fine grained sandstone, with subordinate interbedded siltstone and claystone, which ranges from 1.0 to 3.5 m in thickness; this unit is here informally called the Cedar Wash unit. The basal contact of the Cedar Wash unit is a local unconformity that cuts progressively down across the upper 2.5 m of the parallel-bedded siltstone as it is traced from south to north along Cedar Wash. At the top the Cedar Wash unit is in apparently conformable contact with the lower massive sandstone, although very locally, uppermost Cedar Wash beds have been scoured slightly immediately prior to and during deposition of the basal bed of the lower massive sandstone.

The Cedar Wash unit thickens from south to north by just the amount that beds of the underlying parallel-bedded siltstone are cut out by the unconformity at the base of the Cedar Wash unit (Figure 4). In the southerly exposures the Cedar Wash unit is parallel bedded, but between detailed sections K and L (Figure 4), almost the entire unit passes abruptly into a foreset bedded sequence that is truncated at the top by a very thin topset sequence. The contact between the topset and foreset beds descends gradually to the north and then drops abruptly near detailed section A (Figure 4) to about the middle of the Cedar Wash unit. North of section A the Cedar Wash unit is again parallel bedded but is about 3 times thicker than in the southerly exposures. We interpret the Cedar Wash unit as an estuarine deposit and the foreset beds as a small delta built out from the margin of the depression formed by pre–Cedar Wash erosion.

The Gray Mountain magnetozone occurs entirely within the Cedar Wash unit at Gray Mountain. The base of the magnetozone is coincident with the unconformity at the base of the Cedar Wash unit. At six sites where we have sampled across the lower boundary (five of which are shown in Figure 4) the underlying parallel-bedded siltstone is uniformly normally magnetized, and the sandstone of the Cedar Wash unit overlying the boundary is uniformly reversely magnetized. Separation between drill holes straddling the magnetozone boundary at the different sites is 1–5 cm.

The upper boundary of the Gray Mountain magnetozone is coincident with the contact between foreset and topset beds or with the laterally equivalent bedding plane. At 8 of 11 sites where we have sampled across the upper boundary, or up to the boundary, the rocks below the topset beds are uniformly reversely magnetized. However, at the other three sites (shown by arrows in Figure 4) the rocks just below the contact with the topset beds have a complex pattern of characteristic magnetization. Over a vertical distance of 5–30 cm, directions of magnetization vary from reversed to normal, and many directions are intermediate between reversed and normal. Along one foreset bed, at detailed section B (Figure 4), intermediate directions of magnetization were found more than 1 m below the contact with the topset beds.

As the anomalous sites are all within the foreset sequence and are stratigraphically interspersed with other sites in the foreset sequence where the rocks just below the contact are uniformly reversely magnetized, it is clear that the local complex patterns of magnetization at the top of the Gray Mountain magnetozone cannot be a record of variations in the ancient geomagnetic field. In fact, the directions of characteristic magnetization have been found to vary along a single bed. Partial thermal demagnetization at temperatures as high as 650°C reduces the number of intermediate directions of magnetization by about 60% from the 550°C thermal demagnetization step (Figure 3*a*). All intermediate directions of magnetization move toward typical Triassic reversed directions at the higher thermal demagnetization steps. We interpret the anomalous sites as places where relatively strong components of Triassic normal magnetization, carried in diagenetic hematite, are superimposed on the primary reversed magnetization of the Gray Mountain magnetozone.

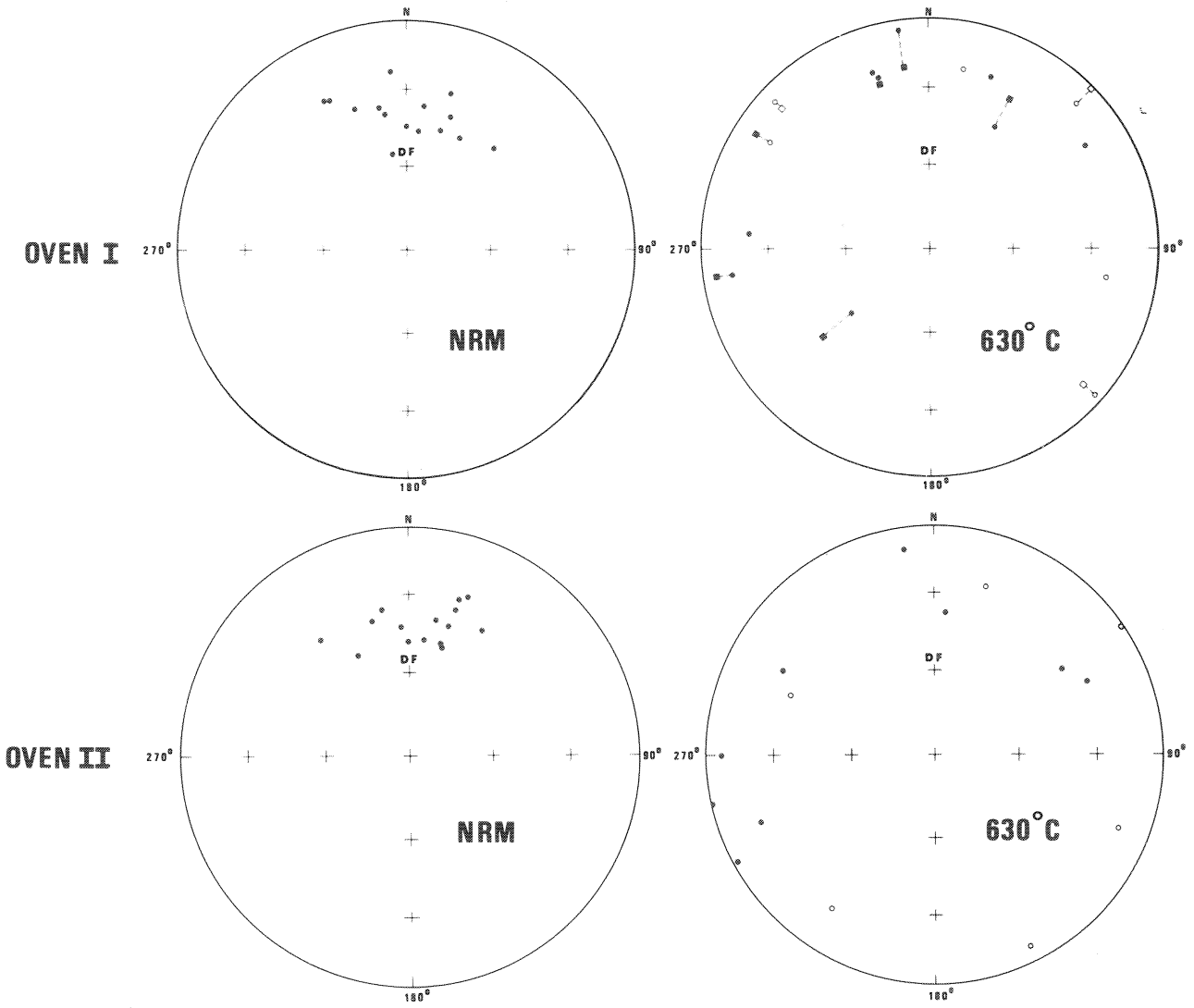


Fig. 5. Equidistant azimuthal projections of directions of magnetization in siltstone and clay stone clasts in the basal 1.5 m of the lower massive sandstone. Squares represent the magnetic directions obtained after the distinctly laminated clasts are rotated to the horizontal around the strike of their laminations. For a description of oven I and oven II, see text. See Figure 3 for an explanation of other symbols.

The topset beds overlying the foreset sequence are everywhere uniformly normally magnetized, as are the succeeding beds of the lower massive sandstone. The topset beds are of diverse lithology, consisting of siltstone and clay stone to the south and sandstone to the north. There is no observed change in lithology at the magnetozone boundary at detailed section A and farther to the north. Thus the upper boundary of the Gray Mountain magnetozone follows a bedding plane but is not lithologically controlled. It is also of interest that the upper boundary of the magnetozone does not follow the most striking lithologic boundary in the section, which is the basal contact of the lower massive sandstone, even though the magnetozone boundary closely underlies this contact.

The precise manner in which the boundaries of the Gray Mountain magnetozone follow stratigraphic contacts indicates that the dominant part of the characteristic magnetization is contemporaneous with deposition of the beds. The relationship is exactly what would be expected if the dominant part of the characteristic magnetization is carried in detrital hematite, as is indicated by petrographic and other evidence [*Elston and Purucker*, 1979]. At a few sites at the upper boundary of the magnetozone the detrital component is subordinate, and an easily recognized secondary component of magnetization of opposite polarity is dominant. Probably there is a subordinate secondary component of magnetization associated with diagenetic hematite pigment in most of the rocks, but it is not easily recognized when it is of the same polarity as the depositional remanent magnetization.

Conglomerate Test

A standard test that allows limits to be placed on the time of acquisition of magnetization is the conglomerate test of *Graham* [1949]. The conglomerate test reveals whether the characteristic magnetization in clasts was acquired before or after their incorporation into a conglomerate.

Siltstone and clay stone clasts occur in the basal 1.5 m of the lower massive sandstone. They are tabular to ovoidal, pebble- to cobble-sized clasts of laminated and massive siltstone and clay stone which are derived from the topset beds of the Cedar Wash unit directly underlying the lower massive sandstone.

The NRM directions of these clasts (Figure 5) and of the

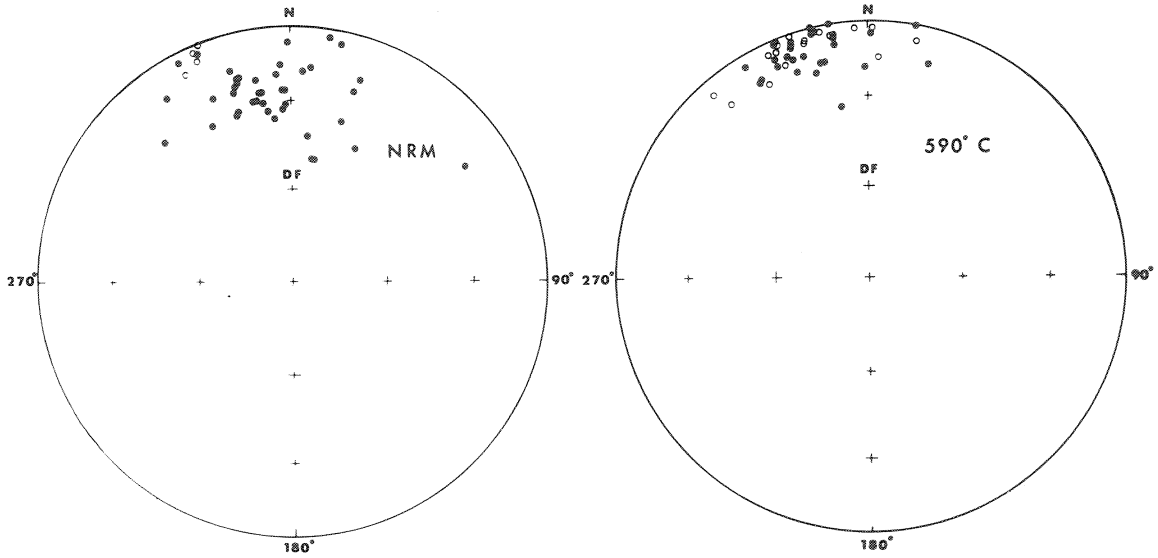


Fig. 6. Equidistant azimuthal projections of directions of magnetization from the lower massive sandstone in the vicinity of siltstone and clay stone clasts. See Figure 3 for an explanation of the symbols.

lower massive sandstone (Figure 6) indicate that both the sandstone and the siltstone contain a major component of present-field magnetization acquired as a consequence of recent weathering. Upon thermal demagnetization, however, completely different relationships are found for the clasts than for the sandstone. The directions of magnetization of the clasts (Figure 5) become scattered with progressive demagnetization, whereas the directions of magnetization of the lower massive sandstone (Figure 6) become clustered about a Triassic normal direction. The scatter of directions in the clasts is not the result of unstable magnetization, because the directions for individual clasts move progressively toward their respective endpoints (Figure 7), with most samples arriving at their stable endpoints at the 630° or 650°C demagnetization steps.

The grouping of directions after partial demagnetization can be tested for randomness by the method of *Watson* [1956], as extended by *Irving* [1964]. By assuming that the points are randomly distributed on the sphere a value R_0 may be calculated which will be exceeded by R (the resultant of the N vectors) with a given probability. A probability value of 0.05 for the threshold of significance is customarily used. The R_0 value for 30 samples at the 5% probability level is 8.80. The value of R for the 30 directions of the clasts at 630°C is 8.95. Hence the distribution of 630°C directions from the clasts is not likely to be taken from a purely random distribution.

By examination of the 630°C directions of magnetization of the clasts it may be seen that there is a preference for the directions to lie near the horizontal plane. This can be accounted for by the tabular shape of the clasts and the ob-

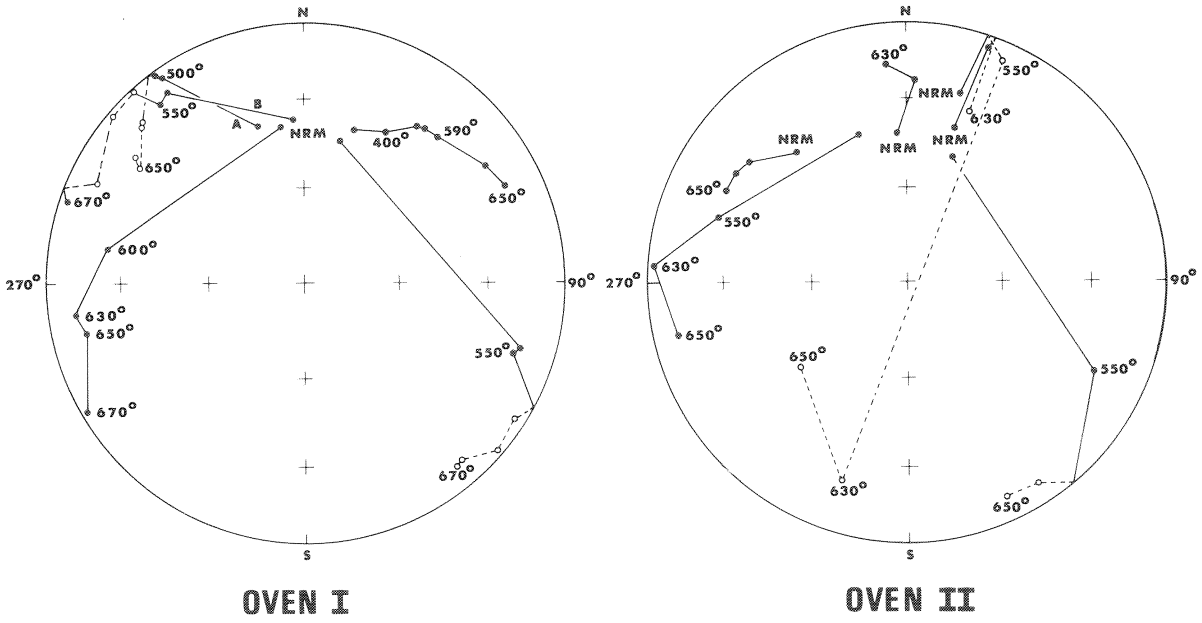


Fig. 7. Demagnetization paths for nine representative siltstone and clay stone clasts from the basal part of the lower massive sandstone. Temperatures are given in degrees Celsius. A and B are the demagnetization paths of two cores taken from the same clast. See Figure 3 for an explanation of other symbols.

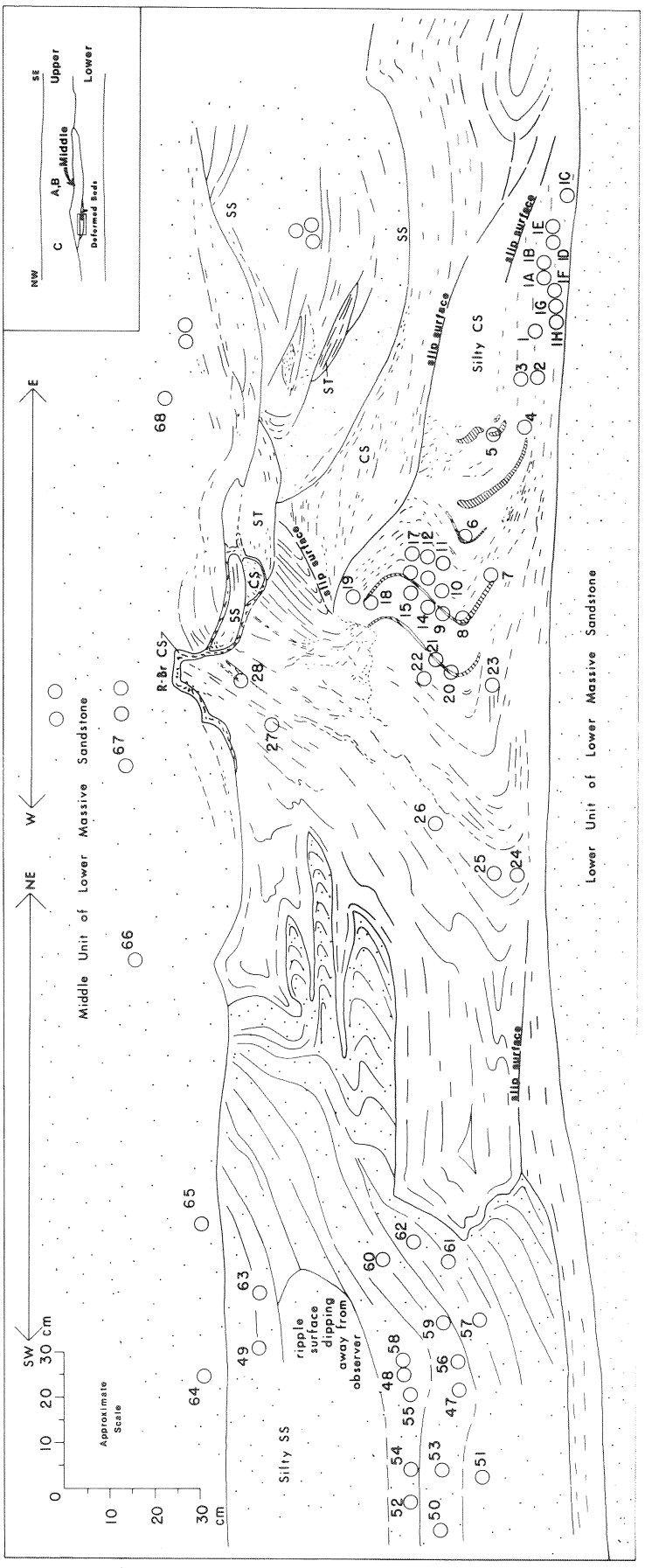


Fig. 8. Sketch of intraformational landslide at the base of the middle unit of the lower massive sandstone. The stratigraphic position of this feature in the lower massive sandstone is shown in the top right-hand corner of the diagram, in which A, B, and C are sites samples for an inclination error test [see *Elston and Purucker*, 1979]. The view of the landslide is a panorama of a near-vertical face, somewhat undercut where silty clay stone is present. Horizontally drilled cores are represented as numbered circles. Their directions of magnetization are shown in Figures 9 and 10. The silty sandstone on the left, in deformational contact with deformed clay stone, is part of a tilted block surrounded by clay stone and is highly deformed at its margins. Beds of the middle unit of the lower massive sandstone overlying the intraformational landslide are undeformed. Beds in the deformed clay stone shown by hatching are local relatively thick green clay stone layers within predominantly dark reddish-brown clay stone. Only the apparent dip is shown in this figure. The true dip was determined after the samples were removed from the outcrop by measuring the orientation of bedding in each sample. CS denotes silty clay stone; ST, siltstone; SS, sandstone, and R-Br, red brown

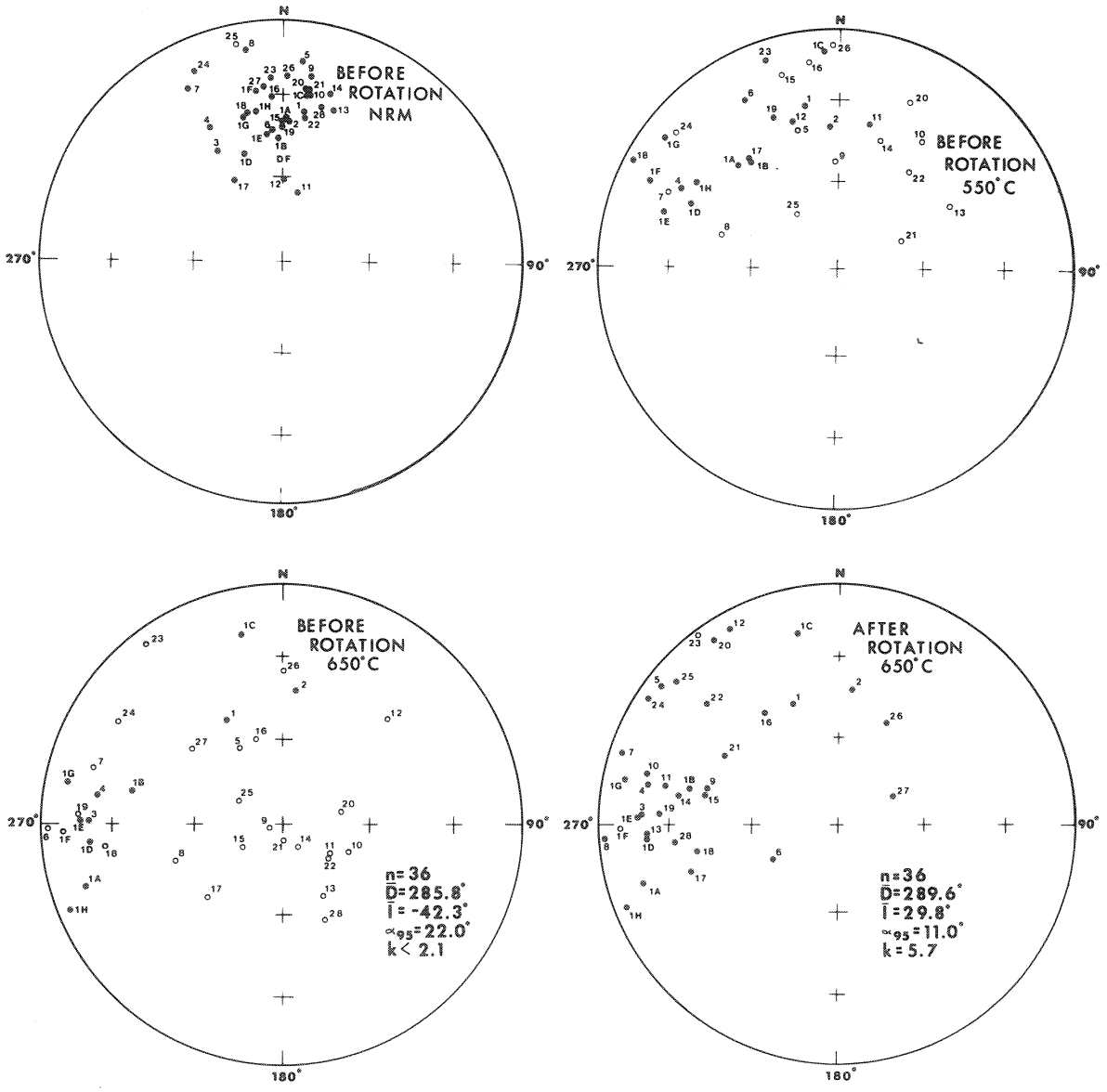


Fig. 9. Equidistant azimuthal projections of directions of magnetization of deformed clay stone before and after structural correction. The numbers refer to core locations shown in Figure 8. See Figure 3 and Table 1 for an explanation of other symbols.

served tendency of the flat sides of the clasts to lie parallel with the bedding of the enclosing sandstone. The flat sides of the tabular clasts also roughly parallel the laminations in the clasts. Nineteen of the 30 clasts have observable laminations, and all of the laminations have dips of less than 25°. Because the mean local inclination of the Triassic magnetic field was approximately 15°, the magnetic vectors in the clasts should lie near the plane of the laminations if the characteristic magnetization of the clasts was set prior to disruption of the source beds. The squares in Figure 5 represent the magnetic inclinations obtained after the distinctly laminated clasts are rotated to the horizontal. Large (20°) rotations in inclination occur in three of the clasts and result in inclinations that lie closer to the mean Triassic inclination.

Another probable source of nonrandomness in the distribution of the directions of magnetization of the clasts is a secondary component of magnetization acquired after deposition of the clasts in the conglomerate. Even though the directions are widely scattered, the mean magnetic declination of the clasts after demagnetization to 630°C is within 4° of the mean characteristic declination for the normal magnetozones at Gray Mountain. As was shown by *Starkey and Palmer* [1970], a secondary component of magnetization as strong as 50% of the total magnetization of the clasts can be present and frequently remain undetected in the test for randomness when $N = 25$. Hence in our test at Gray Mountain, where the secondary component is apparently observable at a level just above the threshold of significance, the mean intensity of the secondary component (at 630°C) may well exceed 50%.

The amount of time that elapsed between deposition of the source beds at the top of the Cedar Wash unit and the disruption of these beds and incorporation of the clasts in the overlying lower massive sandstone was geologically short. No geologists who have examined the basal contact of the lower massive sandstone in detail [*McKee*, 1954; *Stewart et al.*, 1972] have considered that it represents a significant time gap. *Stewart et al.* [1972, p. 23] report that the lower massive sandstone, about 12 km north of the Gray Mountain section, fills small

scours cut into the underlying part of the Moenkopi Formation. In the area studied the contact is essentially planar, and the siltstone pebbles at the base of the lower massive sandstone are the primary physical evidence of a hiatus. The gap in the sedimentary record may be no more than a few thousand years. The primary component of the characteristic magnetization of the clasts was acquired in a time not longer than this interval.

Fold Test

Another test that allows limits to be placed on the time of acquisition of magnetization is the fold test of *Graham* [1949]. The fold test reveals whether the magnetization of folded strata was acquired before or after folding.

Highly deformed silty clay stone and sandstone (Figure 8) occur locally in an intraformational landslide in the middle unit of the lower massive sandstone. The silty clay stone is tightly folded, is in part overturned, and exhibits minor expansion or thickening due to plastic flow along the axial planes of the folds. The axes of the folds strike north-south and are approximately horizontal. Several structural blocks of clay stone separated by slip surfaces are juxtaposed in the landslide. Immediately to the left of the structurally disturbed claystone is a suprajacent block of silty sandstone (Figure 8). At the contact the silty sandstone and clay stone are deformed in a series of isoclinal recumbent folds. The sandstone block, about 2 m across, is surrounded by deformed clay stone. In its interior the block is less deformed and dips fairly consistently to the north. The clay stone appears to have been folded as a consequence of collision of the detached sandstone block with the less competent clay stone. Deformation evidently occurred prior to the complete lithification of either the sandstone or the clay stone. The landslide is overlain directly by an undeformed layer of sandstone of the middle unit of the lower massive sandstone.

The NRM directions for the folded claystone before structural correction (Figure 9, top left) form a cluster that lies between the present field direction and the Triassic normal direction. During stepwise progressive thermal demagnetization to 650°C the directions for the individual specimens moved progressively to the endpoints shown in Figure 9, with intensities of 25% of NRM intensities.

The directions after correction for the observed dips are considerably more clustered than they are before structural correction, suggesting that the magnetization was acquired before folding. A statistical test [*McElhinny,* 1964] can be applied to determine whether a fold test has significance. A test is made of the hypothesis that the values of k_1 and k_2 (the estimates of dispersion before and after structural correction, respectively) are drawn from the same population with dispersion κ. The test reveals that the dispersions k_1 and k_2 are statistically different at greater than the 99.5% confidence level and indicates that the magnetization remaining at 650°C was acquired before folding. Moreover, the magnetically cleaned but structurally uncorrected paleomagnetic directions for the folded clay stone differ markedly from the Triassic normal and reversed directions. The mean inclination after demagnetization to 650°C and before unfolding is −42°. After unfolding, the mean inclination is 30°, having a cone with radius of 11° at the 95% confidence level. The 95% cone of confidence for this inclination lies a few degrees away from the mean normal-polarity inclination of the Moenkopi Formation (Table 1).

The mean magnetic declination after structural correction is approximately 290°. Because this declination is considerably different from either a Triassic normal or a reversed declination, we conclude that the clay stone unit has been rotated azimuthally during folding. Because the mean magnetic inclination after structural correction is positive, as would be expected for beds within the middle normal magnetozone, the clay stone unit evidently is normally magnetized and has been rotated azimuthally about 50° in a counterclockwise direction.

At demagnetization to 550°C the direction of the remanent magnetization of many samples of the deformed clay stone is close to the Triassic normal direction prior to structural correction (Figure 9). This indicates that the magnetization remaining in the samples at the 550°C demagnetization step is dominantly a secondary component acquired after deformation. In most cases this secondary component is removed or greatly reduced by demagnetization to 650°C.

The magnetic directions from the deformed silty sandstone block and from undeformed overlying beds of the middle unit of the lower massive sandstone are shown in Figure 10. Anomalous NRM intensities in these rocks suggested the presence of lightning-induced magnetization, a suggestion that was confirmed by af demagnetization (Figure 10, top right). The IRM apparently was confined to the sandstone, because specimens from the silty clay stone exhibited only minor changes in direction during af demagnetization. After af and subsequent thermal demagnetization, specimens from the undeformed beds overlying the landslide show typical Triassic normal directions, whereas the samples from the deformed sandstone are well grouped with an easterly declination and a moderately steep negative inclination (Figure 10, bottom left). Correction for the dip of the silty sandstone (Figure 10, bottom right) flattens the inclination. Because the directions from the silty sandstone are well grouped and differ substantially from either Triassic normal or reversed directions, we conclude that the deformed sandstone block also has been rotated azimuthally but in a different sense and amount than the silty clay stone.

The low negative inclination after structural correction in the deformed sandstone block probably is best explained in terms of inclination error. If the beds of the sandstone block possessed an original dip in a direction opposite to the Triassic normal field (to the southeast), a depositional remanent magnetic inclination would be expected to be shallower than the Triassic inclination or negative. This exact situation occurs less than 2 m above the deformed sandstone [*Elston and Purucker,* 1979]. Laminated, south-dipping sandstone in the upper unit of the lower massive sandstone here possesses a Triassic normal declination and a low (−6°) negative inclination. If the deformed sandstone acquired its original magnetization in a Triassic normal field and if it had an original magnetization that was steepened (but not reversed) by the same event that caused the azimuthal rotation, then the direction of original dip can be calculated. Such a calculation indicates that beds in the sandstone block originally dipped to the southeast, opposite the Triassic normal-polarity direction. Thus a combination of azimuthal rotation and an inclination error of detrital origin appears to explain the characteristic magnetic direction observed in the sandstone block.

SUMMARY AND DISCUSSION

In rocks of the Moenkopi Formation near Gray Mountain, Arizona, partial thermal and alternating field demagnetiza-

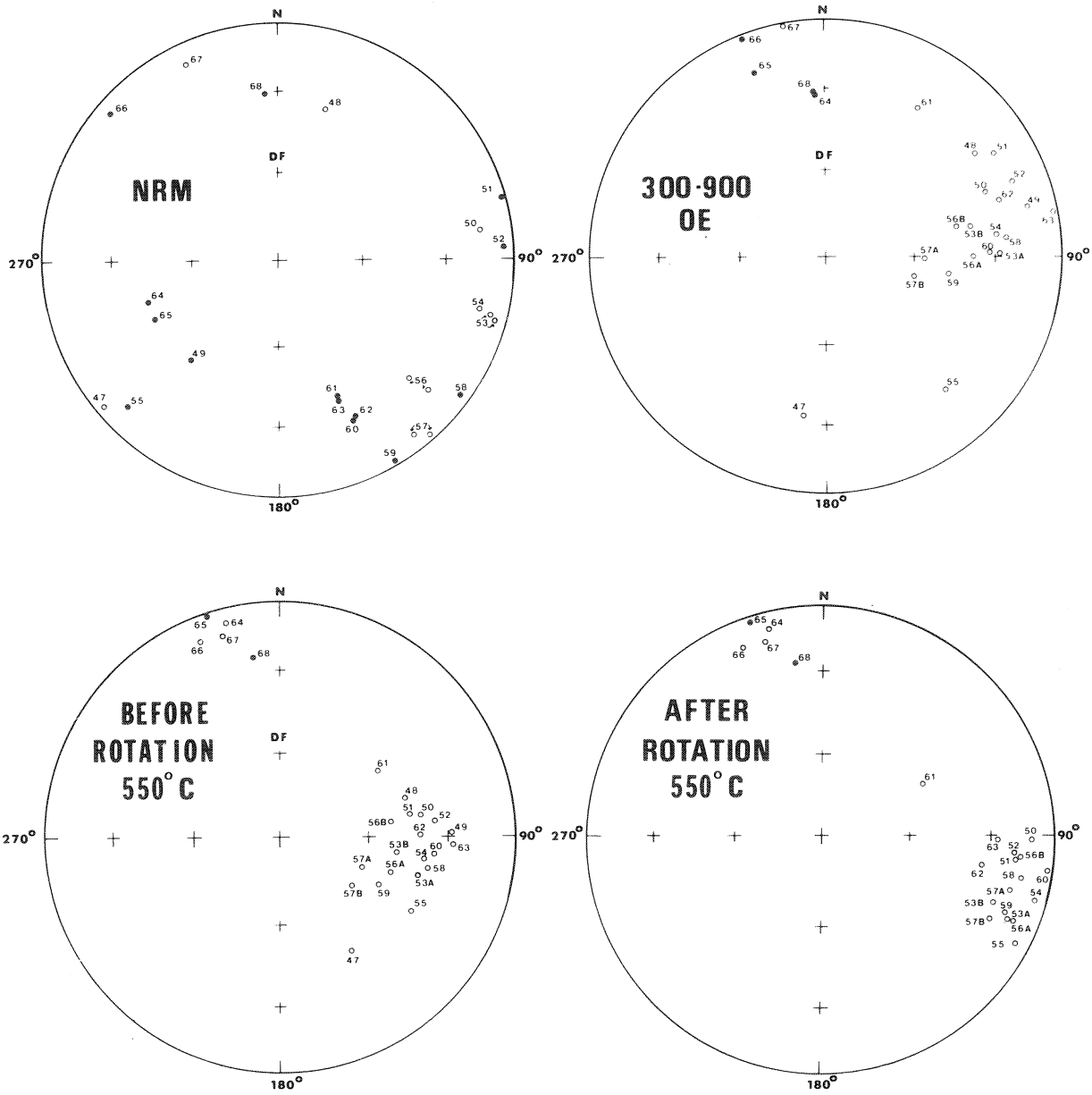


Fig. 10. Equidistant azimuthal projections of the directions of magnetization from the silty sandstone block in the intraformational landslide (Figure 8) and from overlying undeformed sandstone (cores 64–68). The numbers refer to the cores shown in Figure 8. The cores were first subjected to alternating field demagnetization (top right) to remove isothermal components of magnetization before partial thermal demagnetization (bottom left) was carried out. The bottom right-hand plot shows directions from the silty sandstone after correction for dip of the laminations. OE denotes Oersteds. See Figure 3 for explanation of other symbols.

tions remove most of the secondary components of magnetization that have been acquired during recent erosion and weathering. Directions of the remaining (characteristic) magnetization obtained from normally and reversely magnetized rocks are closely similar to characteristic directions reported from the Moenkopi elsewhere on the Colorado Plateau. Two normal and two reversed magnetozones of regional extent are represented in the section at Gray Mountain, and a third normal magnetozone, not found elsewhere, is preserved at the top of the section. The magnetozones are broadly concordant with regional stratigraphic units of the Moenkopi Formation. Detailed study of the lower of the two reversed magnetozones, the Gray Mountain magnetozone, shows that it also is precisely concordant with local stratigraphic units. These relationships, taken together, attest to the general acquisition of a major fraction of the characteristic magnetization across the region during deposition of the formation.

Several lines of evidence indicate that the characteristic magnetization of the Moenkopi rocks commonly, if not generally, comprises at least two components: (1) a primary component that is evidently a depositional remanent magnetization carried in detrital hematite [*Elston and Purucker*, 1979] and (2) a secondary component that evidently is a chemical remanent magnetization carried in diagenetic hematite. The two components are most easily distinguished where they are of opposite polarity, as found locally near the top of the Gray Mountain magnetozone. Sites where a secondary component of opposite polarity dominates the characteristic magnetization are found, with rare possible exceptions, only near magnetozone boundaries. Superposition of a secondary com-

ponent of the same polarity as the primary component, in a given depositional unit, is recognized in special circumstances where the primary component has been rotated structurally, as in the clasts of an intraformational conglomerate and in an intraformational landslide. The occurrence of a secondary magnetization in these diverse circumstances suggests to us that an early secondary chemical remanent magnetization (CRM) is a common component of the characteristic magnetization. Indeed, secondary components of magnetization would be expected to be carried in the nearly ubiquitous earthy hematite that is clearly secondary and that gives the Moenkopi much of its red color [*Walker,* 1976; *Scholle,* 1979; *Elston and Purucker,* 1979]. The extent to which magnetizations carried in earthy hematite contribute to the characteristic magnetization has not been fully resolved. Diagenetic specular hematite also occurs in the Moenkopi rocks at Gray Mountain [*Elston and Purucker,* 1979, Figure 2; *Walker,* 1976; *Scholle,* 1979, p. 125], and this may be the principal carrier of the secondary component of the characteristic magnetization.

The facts that the magnetozone boundaries generally are sharply defined and concordant with the physical stratigraphic units and that only about 4% of the rocks within the magnetozones are anomalously magnetized indicate that secondary components of the characterisitc magnetization either are generally subordinate to the primary component or are of the same polarity and therefore not far separated in time of origin from the primary magnetization. A rough estimate can be made of the average delay between the time of deposition of the sediments and the time of growth through the blocking size of the diagenetic hematite crystals carrying most of the early secondary component of magnetization. This estimate comes from the maximum thickness of beds at conformable magnetozone boundaries in which superposition of two magnetizations of opposite polarity can be recognized. The average thickness found at six complete sections of the Moenkopi in Arizona and southwestern Utah (Figure 1) is ~0.5 m, and the maximum is 2.5 m. Assuming a deposition rate of 3×10^{-5} m yr^{-1} gives an average time delay of the order of 10^4 years and locally as much as $\sim 10^5$ years. This does not mean that earthy hematite could not have crystallized at later times. Considerable growth of hematite crystals after reaching the blocking size may have occurred without changing the CRM. Unresolvable minor components of magnetization, moreover, may be carried in earthy hematite precipitated at much later times. Some rare anomalous directions of characteristic magnetization may be due to such late-crystallizing hematite. Earthy hematite remobilized during recent weathering probably carries most of the magnetization removed by partial thermal demagnetization up to 500°C.

Acknowledgments. We wish to thank the Shoemaker and Elston families for assistance in the collection of samples. S. L. Gillett also assisted in sampling and wrote some valuable computer programs. We would like to thank J. L. Kirschvink for helpful discussions. The paper benefited from reviews by R. Reynolds, S. Grommé, B. Ellwood, and reviewers for the journal.

REFERENCES

Baag, C. G., and C. E. Helsley, Evidence for penecontemporaneous magnetization of the Moenkopi formation, *J. Geophys. Res., 79,* 3308–3320, 1974.

Baldwin, E. J., The Moenkopi Formation of north-central Arizona: An interpretation of ancient environments based upon sedimentary structures, *J. Sediment. Petrology, 43,* 92–106, 1973.

Beck, M. E., Jr., Paleomagnetic and geological implications of magnetic properties of the Triassic diabase of southeastern Pennsylvania, *J. Geophys. Res., 70,* 2845–2856, 1965.

Christensen, F. D., and C. E. Helsley, A paleomagnetic investigation of the Lower Triassic State Bridge Formation in northcentral Colorado (abstract), *Eos Trans. AGU, 55,* 1110, 1974.

Collinson, D. W., Chemical demagnetization, in *Methods in Paleomagnetism,* edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 306–310, Elsevier, New York, 1967.

Collinson, D. W., The role of pigment and specularite in the remanent magnetism of red sandstones, *Geophys. J. Roy. Astron. Soc., 38,* 253–264, 1974.

Cox, A., Anomalous remanent magnetization of basalt, *U.S. Geol. Surv. Bull., 1083E,* 1961.

Doell, R. R., and A. Cox, Measurement of the remanent magnetization of igneous rocks, *U.S. Geol. Surv. Bull., 1203A,* 1965.

Elston, D. P., and M. E. Purucker, Detrital magnetization in red beds of the Moenkopi Formation (Triassic), Gray Mountain, Arizona, *J. Geophys. Res., 84,* 1653–1665, 1979.

Fisher, R. S., Dispersion on a sphere, *Proc. Roy. Soc. London, Ser. A, 217,* 295–305, 1953.

Fuller, M., I. Williams, and K. A. Hoffman, Paleomagnetic records of geomagnetic field reversals and the morphology of the transitional fields, *Rev. Geophys. Space Phys., 17,* 179–203, 1979.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, *J. Geophys. Res., 54,* 131–167, 1949.

Graham, K. W. T., The re-magnetization of a surface outcrop by lightning currents, *Geophys. J., 6,* 85–102, 1961.

Helsley, C. E., Magnetic reversal stratigraphy of the Lower Triassic Moenkopi Formation of western Colorado, *Geol. Soc. Amer. Bull., 80,* 2431–2450, 1969.

Helsley, C., and M. Steiner, Paleomagnetism of the Lower Triassic Moenkopi Formation, *Geol. Soc. Amer. Bull., 85,* 457–464, 1974.

Irving, E., *Paleomagnetism and Its Application to Geological and Geophysical Problems,* 399 pp., John Wiley, New York, 1964.

Irving, E., and M. R. Banks, Paleomagnetic results from the Upper Triassic lavas of Massachusetts, *J. Geophys. Res., 66,* 1935–1939, 1961.

Larochelle, A., and K. L. Currie, Paleomagnetic study of igneous rocks from the Manicouagan structure, Quebec, *J. Geophys. Res., 72,* 4163–4169, 1967.

Larson, E. E., R. P. Hoblitt, and T. R. Walker, Evidence of chemical remagnetization long after deposition in red beds of the Moenkopi Formation near Gray Mountain, Arizona, *Geol. Soc. Amer. Abstr. Programs, 8,* 971, 1976.

McElhinny, M. W., Statistical significance of the fold test in paleomagnetism, *Geophys. J. Roy. Astron. Soc., 8,* 338–340, 1964.

McKee, E. D., Stratigraphy and history of the Moenkopi Formation of Triassic age, *Geol. Soc. Amer. Mem., 61,* 1954.

Opdyke, N. D., The paleomagnetism of the New Jersey Triassic: A field study of the inclination error in red sediments, *J. Geophys. Res., 66,* 1941–1949, 1961.

Purucker, M. E., Magnetic record of lightning strikes in sandstone (abstract), *Eos Trans. AGU, 55,* 1112, 1974.

Robertson, W. A., Manicouagan, Quebec, paleomagnetic results, *Can. J. Earth Sci., 4,* 641–649, 1967.

Roy, J. L., and J. K. Park, Red beds: DRM or CRM?, *Earth Planet. Sci. Lett., 17,* 211–216, 1972.

Scholle, P. A., A color illustrated guide to constituents, textures, and porosities of sandstones and associated rocks, *Amer. Ass. Petrol. Geol. Mem. 28,* 201 pp., 1979.

Shoemaker, E. M., and W. L. Newman, Moenkopi formation (Triassic(?) and Triassic) in salt anticline region, Colorado and Utah, *Amer. Ass. Petrol. Geol. Bull., 43,* 1835–1851, 1959.

Shoemaker, E. M., and M. E. Purucker, 'Gray Mountain' magnetozone in the Moenkopi Formation of Arizona and Utah (abstract), *Eos Trans. AGU, 55*(12), 1108–1109, 1974.

Starkey, J., and H. C. Palmer, The sensitivity of the conglomerate test in paleomagnetism, *Geophys. J. Roy. Astron. Soc., 22,* 235–240, 1970.

Stewart, J. H., F. G. Poole, and R. F. Wilson, Stratigraphy and origin of the Triassic Moenkopi Formation and related strata in the Colorado Plateau region, *U.S. Geol. Surv. Prof. Pap. 691,* 1972.

Walker, T. R., Diagenetic origin of continental red beds, in *The Continental Permian of Central, West and Southern Europe, Proceedings of the NATO Advanced Study Institute,* edited by H. Falke, pp. 240–282, D. Reidel, Hingham, Mass., 1976.

Walker, T. R., and E. E. Larson, Hematite authegenesis in the Moenkopi Formation (Triassic age), Colorado Plateau: A contribution to the origin of magnetism in red beds, *Geol. Soc. Amer. Abstr. Programs, 8,* 1158, 1976.

Ward, L. F., Geology of the Little Colorado River Valley, *Amer. J. Sci., 12,* 401–413, 1901.

Watson, G. S., A test for randomness of directions, *Mon. Notic. Roy. Astron. Soc., Geophys. Suppl., 7,* 160–161, 1956.

(Received November 7, 1977;
revised August 27, 1979;
accepted September 28, 1979).